Exp 15- Filtering Rows with At Least Two Missing (NaN) Values in a Pandas DataFrame

Aim

To write a Pandas program to keep only the rows that contain at least 2 NaN (missing) values in a given DataFrame.

Procedure

1. Import the Pandas library.
2. Create a DataFrame with sample data.
3. Use the `isna()` function to identify the missing values.
4. Count the number of NaN values in each row using `sum(axis=1)`.
5. Filter the rows that contain 2 or more NaN values.
6. Display the resulting DataFrame.

Program

```
import pandas as pd
import numpy as np

# Create the DataFrame

data = {
    'ord_no': [np.nan, np.nan, 70002.0, np.nan, np.nan,
               70005.0, np.nan, 70010.0, 70003.0,
               70012.0, np.nan, np.nan],

    'purch_amt': [np.nan, 270.65, 65.26, np.nan, 948.50,
                  2400.60, 5760.00, 1983.43, 2480.40,
                  250.45, 75.29, np.nan],

    'ord_date': [np.nan, '2012-09-10', np.nan, np.nan,
                 '2012-09-10', '2012-07-27',
                 '2012-09-10', '2012-10-10',
                 '2012-10-10', '2012-06-27',
                 '2012-08-17', np.nan],

    'customer_id': [np.nan, 3001.0, 3001.0, np.nan,
                    3002.0, 3001.0, 3001.0,
                    3004.0, 3003.0, 3002.0,
                    3001.0, np.nan]
}

df = pd.DataFrame(data)

print("Original DataFrame:")
print(df)

# Keep rows with at least 2 NaN values

result = df[df.isna().sum(axis=1) >= 2]

print("\nRows with at least 2 NaN values:")
print(result)
```

OUTPUT

```
Original DataFrame:
     ord_no  purch_amt    ord_date  customer_id
0       NaN        NaN         NaN          NaN
1       NaN     270.65  2012-09-10       3001.0
2   70002.0      65.26         NaN       3001.0
3       NaN        NaN         NaN          NaN
4       NaN     948.50  2012-09-10       3002.0
5   70005.0    2400.60  2012-07-27       3001.0
6       NaN    5760.00  2012-09-10       3001.0
7   70010.0    1983.43  2012-10-10       3004.0
8   70003.0    2480.40  2012-10-10       3003.0
9   70012.0     250.45  2012-06-27       3002.0
10      NaN      75.29  2012-08-17       3001.0
11      NaN        NaN         NaN          NaN

Rows with at least 2 NaN values:
    ord_no  purch_amt ord_date  customer_id
0      NaN        NaN      NaN          NaN
3      NaN        NaN      NaN          NaN
11     NaN        NaN      NaN          NaN
```

Exp 16- Grouping a DataFrame Based on School Code and Checking the Type of GroupBy Object

Aim

To write a Pandas program to split a DataFrame into groups based on the school code and check the type of the GroupBy object.

Procedure

1. Import the Pandas library.
2. Create a DataFrame containing student details.
3. Use the `groupby()` function to group the data based on the `school` column.
4. Display each group separately.
5. Check the type of the GroupBy object using the `type()` function.
6. Print the result.

Program

```
import pandas as pd

# Create the DataFrame

data = {
    'school': ['s001', 's002', 's003', 's001', 's002', 's004'],
    'class': ['V', 'V', 'VI', 'VI', 'V', 'VI'],
    'name': ['Alberto Franco', 'Gino Mcneill', 'Ryan Parkes',
             'Eesha Hinton', 'Gino Mcneill', 'David Parkes'],
    'date_Of_Birth': ['15/05/2002', '17/05/2002', '16/02/1999',
                      '25/09/1998', '11/05/2002', '15/09/1997'],
    'age': [12, 12, 13, 13, 14, 12],
    'height': [173, 192, 186, 167, 151, 159],
    'weight': [35, 32, 33, 30, 31, 32],
    'address': ['street1', 'street2', 'street3',
                'street1', 'street2', 'street4']
}

df = pd.DataFrame(data)

print("Original DataFrame:\n")
print(df)

# Group by school code

grouped = df.groupby('school')

print("\nGroups based on school code:\n")

for school, group in grouped:
    print("School Code:", school)
    print(group)
    print()

# Check the type of GroupBy object

print("Type of GroupBy object:")
print(type(grouped))
```

OUTPUT

```
Original DataFrame:

  school class            name date_Of_Birth  age  height  weight  address
0   s001     V  Alberto Franco    15/05/2002   12     173      35  street1
1   s002     V    Gino Mcneill    17/05/2002   12     192      32  street2
2   s003    VI     Ryan Parkes    16/02/1999   13     186      33  street3
3   s001    VI    Eesha Hinton    25/09/1998   13     167      30  street1
4   s002     V    Gino Mcneill    11/05/2002   14     151      31  street2
5   s004    VI    David Parkes    15/09/1997   12     159      32  street4

Groups based on school code:

School Code: s001
  school class            name date_Of_Birth  age  height  weight  address
0   s001     V  Alberto Franco    15/05/2002   12     173      35  street1
3   s001    VI    Eesha Hinton    25/09/1998   13     167      30  street1

School Code: s002
  school class          name date_Of_Birth  age  height  weight  address
1   s002     V  Gino Mcneill    17/05/2002   12     192      32  street2
4   s002     V  Gino Mcneill    11/05/2002   14     151      31  street2

School Code: s003
  school class         name date_Of_Birth  age  height  weight  address
2   s003    VI  Ryan Parkes    16/02/1999   13     186      33  street3

School Code: s004
  school class          name date_Of_Birth  age  height  weight  address
5   s004    VI  David Parkes    15/09/1997   12     159      32  street4

Type of GroupBy object:
<class 'pandas.core.groupby.generic.DataFrameGroupBy'>
```

Exp 17-Grouping a DataFrame by School Code and Finding Mean, Minimum, and Maximum Age

Aim

To write a Pandas program to split the DataFrame based on school code and find the mean, minimum, and maximum values of age for each school.

Procedure

1. Import the Pandas library.
2. Create a DataFrame containing student details.
3. Group the DataFrame using the `school` column with the `groupby()` function.
4. Select the `age` column from the grouped data.
5. Use the `agg()` function to calculate the mean, minimum, and maximum ages.
6. Display the result.

Program

```
import pandas as pd

# Create the DataFrame

data = {
    'school': ['s001', 's002', 's003', 's001', 's002', 's004'],
    'class': ['V', 'V', 'VI', 'VI', 'V', 'VI'],
    'name': ['Alberto Franco', 'Gino Mcneill', 'Ryan Parkes',
             'Eesha Hinton', 'Gino Mcneill', 'David Parkes'],
    'date_Of_Birth': ['15/05/2002', '17/05/2002', '16/02/1999',
                      '25/09/1998', '11/05/2002', '15/09/1997'],
    'age': [12, 12, 13, 13, 14, 12],
    'height': [173, 192, 186, 167, 151, 159],
    'weight': [35, 32, 33, 30, 31, 32],
    'address': ['street1', 'street2', 'street3',
                'street1', 'street2', 'street4']
}

df = pd.DataFrame(data)

# Group by school and calculate statistics

result = df.groupby('school')['age'].agg(['mean', 'min', 'max'])

print(result)
```

OUTPUT

```
        mean  min  max
school
s001    12.5   12   13
s002    13.0   12   14
s003    13.0   13   13
s004    12.0   12   12
```